ELEC-E8126 – Robotic manipulation

Exercise 3 optional assignment
23.2.2020
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The number of object and scene features are different: there are 2202 object features and 307200 scene features. The numbers were obtained by printing the sizes of their corresponding arrays. One reason for this difference is that the scene is significantly larger than the manipulated object. Assuming a relatively uniform sampling density, a larger target will have more samples taken overall.

A minimum of three sampled points are required to generate or prereject a pose in 3D space. Since individual points do not contain direct information about the rotations of an object, having fewer than three points will leave some degrees of freedom for the pose of an object. In 3D space, one sampled point fixes the pose to rotate freely around that point. Two points will restrict the pose to rotate around a line. The third point will then define the rotation around the line defined by the first two points. This is visualized in the following figure.

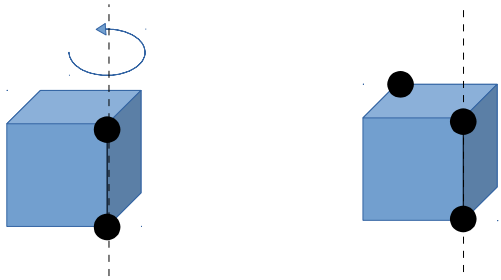

Figure 1: two points restrict movement to rotate around a line (left), the third fixes the position

The program given in the exercise was run 10 times. The inlier fractions, alignment result, and alignment times for the runs are listed below.

Run	Inlier fraction	Alignment time (ms)	Alignment successful
1	$\frac{737}{2202}$	8396	Yes
2	$\frac{737}{2202}$	7967	Yes
3	$\frac{737}{2202}$	7990	Yes
4	$\frac{737}{2202}$	8138	Yes
5	$\frac{737}{2202}$	7885	Yes
6	$\frac{737}{2202}$	7444	Yes
7	$\frac{737}{2202}$	7897	Yes
8	$\frac{737}{2202}$	7727	Yes

9	$\frac{737}{2202}$	8234	Yes
10	$\frac{737}{2202}$	7635	Yes

The minimum alignment time was 7444 ms, the maximum 8396 ms, and the average was 7931.3 ms. The alignment varies because the RANSAC algorithm, which is used to find the inliers and outliers from a dataset, does not always perform the same number of iterations. The program does set a maximum number of iterations for the algorithm, but the execution may stop before reaching that value if a satisfying result is found before that.

The alignment was successful every time.

The transformations given by the pose estimation are listed in the following table.

Run	Translation	Rotation matrix	Euler angles
1	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
2	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
3	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
4	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
5	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
6	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
7	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
8	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$
9	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$

10	$\begin{bmatrix}0.114 & -0.061 & 1.496\end{bmatrix}^\top$	$\begin{bmatrix}0.999 & -0.005 & -0.046\\ -0.035 & -0.736 & -0.676\\ -0.030 & 0.677 & -0.735\end{bmatrix}$	$x: 2.3980$ $y: -0.0460$ $z: 0.0050$

The inlier fraction that the program prints is 737/2202, which is approximately 0.335. The inlier fraction describes the threshold for accepting a hypothesis for alignment, i.e. the ratio of correctly aligned points to the total number of object points is high enough. The inlier fraction would be difficult to set close to 1 because in a realistic case, some of the object points may not be visible, thus requiring nearly all points to be correctly aligned would be infeasible if not impossible.

The positions and orientations were the same for each run. The pose of the obtained frame is plotted in Matlab using plotTransforms (figure 2). The used script is shown at the end of this document.

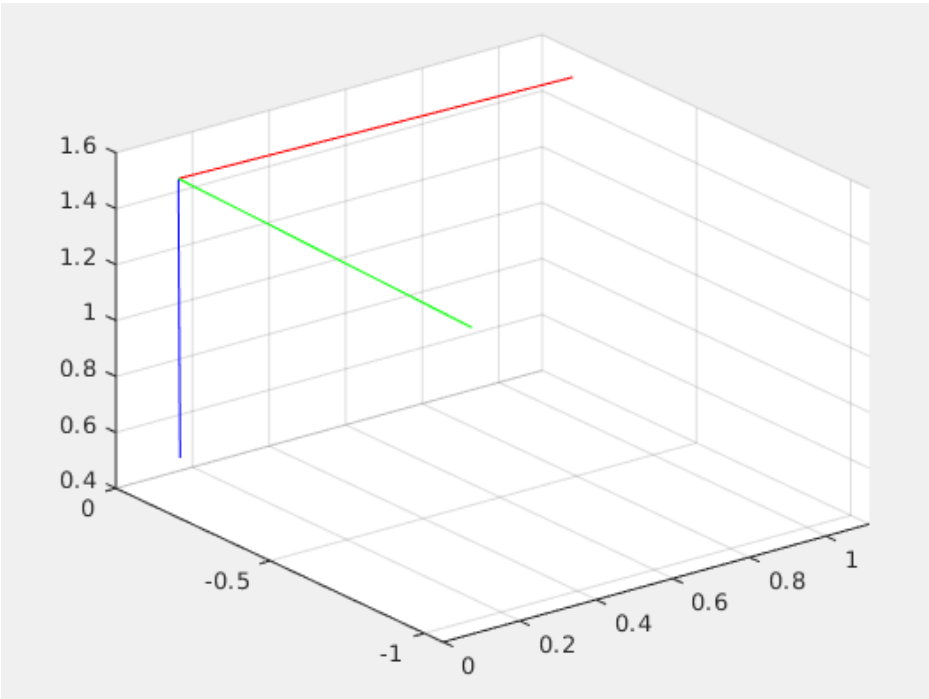

Figure 2: Pose of the frame obtained by the basic alignment

The ICP algorithm was implemented and the program was run another 10 times. The obtained results are shown in the following table.

Run	Translation	Rotation matrix	Euler angles
1	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
2	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
3	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$

4	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
5	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
6	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
7	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
8	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
9	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$
10	$\begin{bmatrix}-0.028 & -0.189 & 0.843\end{bmatrix}^\top$	$\begin{bmatrix}0.858 & 0.130 & 0.497\\ -0.035 & 0.609 & 0.643\\ -0.030 & -0.782 & 0.583\end{bmatrix}$	$x: -0.8343$ $y: 0.5200$ $z: -0.1504$

Again, the orientations and translations were the same between runs. The pose of the obtained coordinate frame is shown in figure 3.

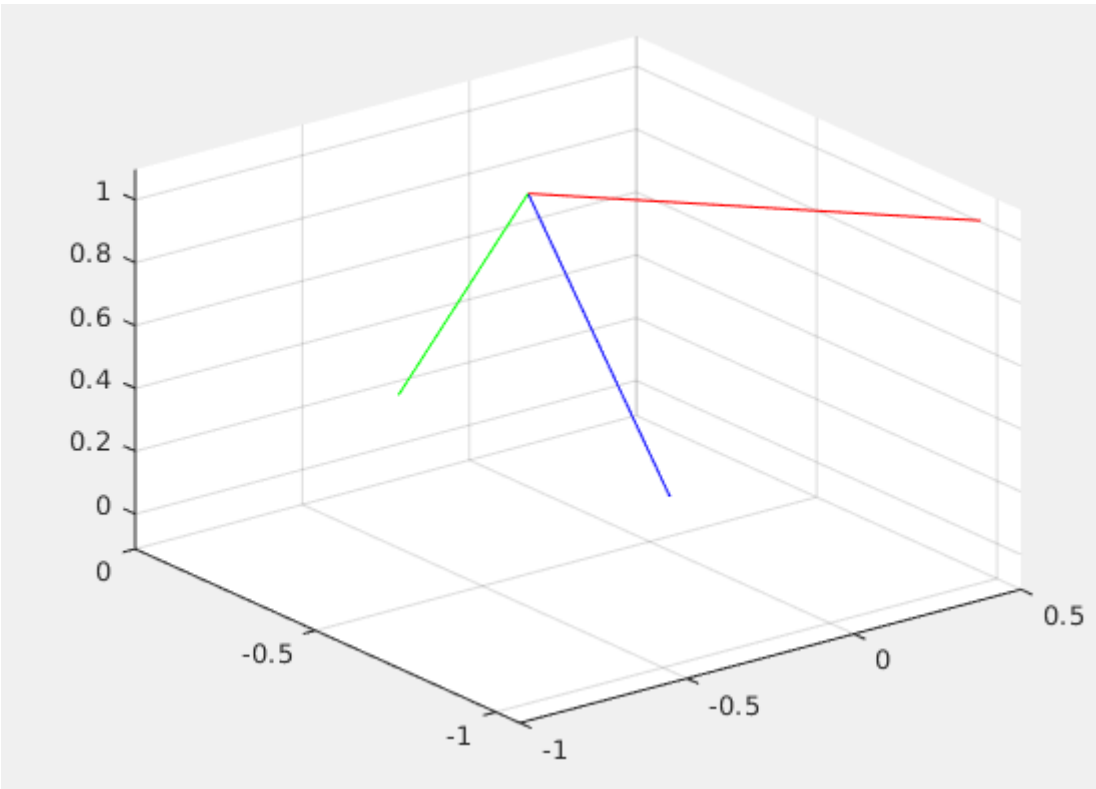

Figure 3: Pose of the frame obtained with ICP

Compared to the results obtained with the basic alignment, the ICP algorithm produced more significant rotations and a larger translation from the origin.

The iterative closest point algorithm can be used to fine-tune an alignment towards some local minimum in the solution space. In the case of misalignment, the basic alignment will have produced something that is in the general vicinity of the solution, i.e. the global minimum. In this case, the ICP algorithm will be of use, since it will bring the alignment more accurately to the local minimum, thus improving the overall alignment. In the case of total failure, the basic alignment has not converged at all, thus the ICP algorithm will be unable to converge to a solution. A note about the ICP algorithm: even if the algorithm converges, it is not guaranteed to have found the true global minimum, only a local one.

Appendix

A Matlab script for plotting poses

```
T = [0.114 -0.061 1.496];
R_mat = [0.999 -0.005 -0.046 ; -0.035 -0.736 -0.676 ; -0.030 0.677 -0.735];
R = [0 rotm2eul(R_mat, "XYZ")]
T_icp = [-0.028 -0.189 0.843];
R_icp_mat = [0.858 0.130 0.497 ; -0.035 0.609 0.643 ; -0.030 -0.782 0.583];
R_icp = [0 rotm2eul(R_icp_mat, "XYZ")]

grid on; hold on;
plotTransforms(T, R);
figure;
grid on; hold on;
plotTransforms(T_icp, R_icp);
```